

JRWatch — Schematic Reference (netlist listing)

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C1 — 10uF

Capacitor_SMD:C_0603_1608Metric LCSC C19702 MPN CL10A106KP8NNNC

Pad	Net
1	VBAT
2	GND

C2 — 10uF

Capacitor_SMD:C_0603_1608Metric LCSC C19702 MPN CL10A106KP8NNNC

Pad	Net
1	VSYS
2	GND

C3 — 2.2uF

Capacitor_SMD:C_0603_1608Metric LCSC C23630 MPN CL10B225KO8NNNC

Pad	Net
1	VSYS
2	GND

C4 — 100nF

Capacitor_SMD:C_0402_1005Metric LCSC C1525 MPN CL05B104KO5NNNC

Pad	Net
1	VSYS
2	GND

C5 — 10uF

Capacitor_SMD:C_0603_1608Metric LCSC C19702 MPN CL10A106KP8NNNC

Pad	Net
1	VBUS_USB
2	GND

C6 — 1uF

Capacitor_SMD:C_0402_1005Metric LCSC C52923 MPN CL05A105KA5NQNC

Pad	Net
1	VBUS_OUT
2	GND

C7 — 10uF

Capacitor_SMD:C_0603_1608Metric LCSC C19702 MPN CL10A106KP8NNNC

Pad	Net
1	1V8_AUX
2	GND

C8 — 10uF

Capacitor_SMD:C_0603_1608Metric LCSC C19702 MPN CL10A106KP8NNNC

Pad	Net
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1	3V0
2	GND

C9 — 1uF

Capacitor_SMD:C_0402_1005Metric LCSC C52923 MPN CL05A105KA5NQNC

Pad	Net
1	VDD_DISP
2	GND

C10 — 1uF

Capacitor_SMD:C_0402_1005Metric LCSC C52923 MPN CL05A105KA5NQNC

Pad	Net
1	VDD_IMU
2	GND

C11 — 100nF

Capacitor_SMD:C_0402_1005Metric LCSC C1525 MPN CL05B104KO5NNNC

Pad	Net
1	3V0
2	GND

C12 — 10uF

Capacitor_SMD:C_0603_1608Metric LCSC C19702 MPN CL10A106KP8NNNC

Pad	Net
1	3V0
2	GND

C13 — 100nF

Capacitor_SMD:C_0402_1005Metric LCSC C1525 MPN CL05B104KO5NNNC

Pad	Net
1	3V0
2	GND

C14 — 100nF

Capacitor_SMD:C_0402_1005Metric LCSC C1525 MPN CL05B104KO5NNNC

Pad	Net
1	3V0
2	GND

C15 — 100nF

Capacitor_SMD:C_0402_1005Metric LCSC C1525 MPN CL05B104KO5NNNC

Pad	Net
1	VBUS_OUT
2	GND

C16 — 12pF

Capacitor_SMD:C_0402_1005Metric LCSC C1546 MPN CL05C120JB5NNNC

Pad	Net
1	XL1
2	GND

C17 — 12pF

Capacitor_SMD:C_0402_1005Metric LCSC C1546 MPN CL05C120JB5NNNC

Pad	Net
1	XL2
2	GND

C18 — 100nF

Capacitor_SMD:C_0402_1005Metric LCSC C1525 MPN CL05B104KO5NNNC

Pad	Net
1	VDD_IMU
2	GND

C19 — 100nF

Capacitor_SMD:C_0402_1005Metric LCSC C1525 MPN CL05B104KO5NNNC

Pad	Net
1	3V0
2	GND

C20 — 1uF

Capacitor_SMD:C_0402_1005Metric LCSC C52923 MPN CL05A105KA5NQNC

Pad	Net
1	VDD_DISP
2	GND

C21 — 100nF

Capacitor_SMD:C_0402_1005Metric LCSC C1525 MPN CL05B104KO5NNNC

Pad	Net
1	VDD_DISP
2	GND

C22 — 1uF

Capacitor_SMD:C_0402_1005Metric LCSC C52923 MPN CL05A105KA5NQNC

Pad	Net
1	VDD_DISP
2	GND

C23 — 100nF

Capacitor_SMD:C_0402_1005Metric LCSC C1525 MPN CL05B104KO5NNNC

Pad	Net
1	VDD_DISP
2	GND

D1 — ESD9B5.0

Diode_SMD:D_SOD-923 LCSC C3008065 MPN ESD9B5.0ST5G

Pad	Net
1	BTN1_PAD
2	GND

D2 — ESD9B3.3

Diode_SMD:D_SOD-923 LCSC C96512 MPN ESD9B3.3ST5G

Pad	Net
1	BTN2_PAD

2	GND
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J1 — TYPE-C-31-M-12

Connector_USB:USB_C_Receptacle_HRO_TYPE-C-31-M-12 LCSC C165948 MPN TYPE-C-31-M-12

Pad	Net
A1	GND
A12	GND
A4	VBUS_USB
A5	CC1
A6	USB_DP_CONN
A7	USB_DM_CONN
A9	VBUS_USB
B1	GND
B12	GND
B4	VBUS_USB
B5	CC2
B6	USB_DP_CONN
B7	USB_DM_CONN
B9	VBUS_USB
SH	GND

J2 — FH12A-10S-0.5SH

Connector_FFC-FPC:Hirose_FH12-10S-0.5SH_1x10-1MP_P0.50mm_Horizontal LCSC C5139870 MPN FH12A-10S-0.5SH(55)

Pad	Net
1	DISP_SCK
2	DISP_MOSI
3	DISP_CS
4	EXTCOMIN
5	DISP_ON
6	VDD_DISP
7	VDD_DISP
8	VDD_DISP
9	GND
10	GND
MP	GND

J3 — SM02B-SRSS-TB

Connector_JST:JST_SH_SM02B-SRSS-TB_1x02-1MP_P1.00mm_Horizontal LCSC C160402 MPN SM02B-SRSS-TB(LF)(SN)

Pad	Net
1	VBAT
2	GND
MP	GND

J4 — TC2030-IDC-NL

Connector:Tag-Connect_TC2030-IDC-NL_2x03_P1.27mm_Vertical MPN TC2030-IDC-NL

Pad	Net
1	3V0
2	SWDIO
3	nRESET
4	SWDCLK
5	GND
6	SWO

L1 — 2.2uH

JRWATCH:L_Murata_DFE201612E_2016 LCSC C337893 MPN DFE201612E-2R2M=P2

Pad	Net
1	N\$1

2	1V8_AUX
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L2 — 2.2uH

JRWatch:L_Murata_DFE201612E_2016 LCSC C337893 MPN DFE201612E-2R2M=P2

Pad	Net
1	N\$3
2	3V0

R1 — 47k

Resistor_SMD:R_0402_1005Metric LCSC C25563 MPN 0402WGF4702TCE

Pad	Net
1	N\$2
2	GND

RT1 — 10k NTC

Resistor_SMD:R_0402_1005Metric LCSC C77131 MPN NCP15XH103F03RC

Pad	Net
1	NTC
2	GND

R2 — 150k

Resistor_SMD:R_0402_1005Metric LCSC C25867 MPN 0402WGF1503TCE

Pad	Net
1	N\$4
2	GND

R3 — 4.7k

Resistor_SMD:R_0402_1005Metric LCSC C25900 MPN 0402WGF4701TCE

Pad	Net
1	I2C_SDA
2	3V0

R4 — 4.7k

Resistor_SMD:R_0402_1005Metric LCSC C25900 MPN 0402WGF4701TCE

Pad	Net
1	I2C_SCL
2	3V0

R5 — 100k

Resistor_SMD:R_0402_1005Metric LCSC C25741 MPN 0402WGF1003TCE

Pad	Net
1	IMU_CS
2	3V0

R6 — 100k

Resistor_SMD:R_0402_1005Metric LCSC C25741 MPN 0402WGF1003TCE

Pad	Net
1	DISP_CS
2	GND

R7 — 100R

Resistor_SMD:R_0402_1005Metric LCSC C25076 MPN 0402WGF1000TCE

Pad	Net
1	SHPHLD
2	BTN1_PAD

R8 — 100R

Resistor_SMD:R_0402_1005Metric LCSC C25076 MPN 0402WGF1000TCE

Pad	Net
1	BTN2
2	BTN2_PAD

SW1 — PWR/SHIP

Button_Switch_SMD:SW_Push_1P1T-MP_NO_Horizontal_Alps_SKRTLAE010 LCSC C110293 MPN SKRTLAE010

Pad	Net
1	BTN1_PAD
2	GND
MP	GND

SW2 — USER

Button_Switch_SMD:SW_Push_1P1T-MP_NO_Horizontal_Alps_SKRTLAE010 LCSC C110293 MPN SKRTLAE010

Pad	Net
1	BTN2_PAD
2	GND
MP	GND

U1 — MDBT50Q-1MV2

RF_Module:Raytac_MDBT50Q LCSC C5118826 MPN MDBT50Q-1MV2

Pad	Net
1	GND
2	GND
6	BTN2
15	GND
17	XL1
18	XL2
28	3V0
29	PMIC_INT
30	3V0
32	VBUS_OUT
33	GND
34	USB_DM
35	USB_DP
36	I2C_SDA
37	I2C_SCL
38	DISP_ON
39	DISP_CS
40	nRESET
41	DISP_SCK
42	IMU_SCK
43	IMU_MOSI
44	DISP_MOSI
45	IMU_MISO
46	IMU_CS
47	SWO
49	IMU_INT1
50	EXTCOMIN
51	SWDIO
53	SWDCLK
55	GND

U2 — nPM1300-QEAA-R7

Package_DFN_QFN:VQFN-32-1EP_5x5mm_P0.5mm_EP3.5x3.5mm_ThermalVias LCSC C7501206 MPN nPM1300-QEAA-R7

Pad	Net
1	1V8_AUX
2	GND
3	N\$1
4	VSYS
5	N\$3
6	GND
7	PMIC_INT
12	3V0
13	I2C_SDA
14	I2C_SCL
15	SHPHLD
16	N\$4
17	N\$2
18	NTC
19	VBAT
20	VSYS
21	VBUS_USB
22	VBUS_OUT
23	CC1
24	CC2
28	3V0
29	VDD_DISP
30	3V0
31	VDD_IMU
32	3V0
33	GND

U3 — BMI270

Package_LGA:Bosch_LGA-14_3x2.5mm_P0.5mm LCSC C2836813 MPN BMI270

Pad	Net
1	IMU_MISO
2	3V0
3	3V0
4	IMU_INT1
5	3V0
6	GND
7	GND
8	VDD_IMU
12	IMU_CS
13	IMU_SCK
14	IMU_MOSI

U4 — USBLC6-2SC6

Package_TO_SOT_SMD:SOT-23-6 LCSC C7519 MPN USBLC6-2SC6

Pad	Net
1	USB_DM_CONN
2	GND
3	USB_DP_CONN
4	USB_DP
5	VBUS_USB
6	USB_DM

Y1 — 32.768kHz

Crystal:Crystal_SMD_3215-2Pin_3.2x1.5mm LCSC C32346 MPN Q13FC13500004

Pad	Net
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1	XL2
2	XL1

Net index

Net	Pins
1V8_AUX	C7.1, L1.2, U2.1
3V0	C11.1, C12.1, C13.1, C14.1, C19.1, C8.1, J4.1, L2.2, R3.2, R4.2, R5.2, U1.28, U1.30, U2.12, U2.28, U2.30, U2.32, U3.2, U3.3, U3.5
BTN1_PAD	D1.1, R7.2, SW1.1
BTN2	R8.1, U1.6
BTN2_PAD	D2.1, R8.2, SW2.1
CC1	J1.A5, U2.23
CC2	J1.B5, U2.24
DISP_CS	J2.3, R6.1, U1.39
DISP_MOSI	J2.2, U1.44
DISP_ON	J2.5, U1.38
DISP_SCK	J2.1, U1.41
EXTCOMIN	J2.4, U1.50
GND	C1.2, C10.2, C11.2, C12.2, C13.2, C14.2, C15.2, C16.2, C17.2, C18.2, C19.2, C2.2, C20.2, C21.2, C22.2, C23.2, C3.2, C4.2, C5.2, C6.2, C7.2, C8.2, C9.2, D1.2, D2.2, J1.A1, J1.A12, J1.B1, J1.B12, J1.SH, J2.10, J2.9, J2.MP, J3.2, J3.MP, J4.5, R1.2, R2.2, R6.2, RT1.2, SW1.2, SW1.MP, SW2.2, SW2.MP, U1.1, U1.15, U1.2, U1.33, U1.55, U2.2, U2.33, U2.6, U3.6, U3.7, U4.2
I2C_SCL	R4.1, U1.37, U2.14
I2C_SDA	R3.1, U1.36, U2.13
IMU_CS	R5.1, U1.46, U3.12
IMU_INT1	U1.49, U3.4
IMU_MISO	U1.45, U3.1
IMU_MOSI	U1.43, U3.14
IMU_SCK	U1.42, U3.13
N\$1	L1.1, U2.3
N\$2	R1.1, U2.17
N\$3	L2.1, U2.5
N\$4	R2.1, U2.16
NTC	RT1.1, U2.18
PMIC_INT	U1.29, U2.7
SHPHLD	R7.1, U2.15
SWDCLK	J4.4, U1.53
SWDIO	J4.2, U1.51
SWO	J4.6, U1.47
USB_DM	U1.34, U4.6
USB_DM_CONN	J1.A7, J1.B7, U4.1
USB_DP	U1.35, U4.4
USB_DP_CONN	J1.A6, J1.B6, U4.3
VBAT	C1.1, J3.1, U2.19
VBUS_OUT	C15.1, C6.1, U1.32, U2.22
VBUS_USB	C5.1, J1.A4, J1.A9, J1.B4, J1.B9, U2.21, U4.5
VDD_DISP	C20.1, C21.1, C22.1, C23.1, C9.1, J2.6, J2.7, J2.8, U2.29
VDD_IMU	C10.1, C18.1, U2.31, U3.8
VSYS	C2.1, C3.1, C4.1, U2.20, U2.4
XL1	C16.1, U1.17, Y1.2
XL2	C17.1, U1.18, Y1.1
nRESET	J4.3, U1.40